Review Questions Answers

1. The most basic circuitry-level computer language is _____.

    a. machine language

    b. Java

    c. high-level language

    d. C++

    Answer: a

    Feedback:

    The most basic circuitry-level computer language is machine language. Machine language, or machine code, is the most basic set of instructions a computer can execute. Java and C++ are both high-level languages and are the opposite of circuitry-level computer language.

2. Languages that let you use an easily understood vocabulary of descriptive terms, such as *read*, *write*, or *add*, are known as _____languages.

    a. procedural

    b. high-level

    c. machine

    d. object-oriented

    Answer: b

    Feedback:

    High-level languages use English-like terms; Java is an example of a high-level language. Procedural languages are those that run by executing a series of procedures or methods. Machine-level languages do not use English-like terms; they use 1s and 0s. Object-oriented languages are run by declaring and using objects that contain data and methods.

3. The rules of a programming language constitute its _____.

    a. syntax

    b. logic

c. format

d. objects

Answer: a

Feedback:

The rules of a programming language constitute its syntax.

4. A _____ translates high-level language statements into machine code.

a. programmer

b. syntax detector

c. compiler

d. decipherer

Answer: c

Feedback:

A compiler translates high-level language statements into machine code. A programmer writes high-level language statements but does not translate them. “Syntax detector” and “decipherer” are not terms used in programming.

5. Named computer memory locations are called _____.

a. compilers

b. variables

c. addresses

d. appellations

Answer: b

Feedback:

Named computer memory locations are variables. Compilers translate programming statements into machine language; they are not memory locations. Addresses are unnamed computer memory locations. “Appellations” is not a term used in programming.

6. The individual operations used in a computer program are often grouped into logical units called

_____.

a. procedures

b. variables

c. constants

d. logistics

Answer: a

Feedback:

The individual operations used in a computer program are often grouped into logical units called procedures. Variables are named memory locations, and constants are values that do not change; they are not groups of logical operations. "Logistics" is not a term commonly used in programming.

7. Envisioning program components as objects that are similar to concrete objects in the real world is the hallmark of _____.

a. command-line operating systems

b. procedural programming

c. object-oriented programming

d. machine languages

Answer: c

Feedback:

Envisioning program components as objects that are similar to concrete objects in the real world is the hallmark of object-oriented programming.

8. The values of an object's attributes are known as its _____.

a. state

b. orientation

c. methods

d. condition

Answer: a

Feedback:

The values of an object's attributes are known as its state.

9. An instance of a class is a(n) _____.

    a. method

    b. procedure

    c. object

    d. case

    Answer: c

    Feedback:

    An instance of a class is an object.

10. Java is architecturally _____.

    a. neutral

    b. oriented

    c. specific

    d. abstract

    Answer: a

    Feedback:

    Java is architecturally neutral.

11. You must compile classes written in Java into _____.

    a. bytecode

    b. source code

    c. Javadoc statements

    d. object code

    Answer: a

    Feedback:

    You must compile classes written in Java into bytecode. Source code is the high-level programming statements. Javadoc statements are a type of comment used for

documentation. Object code is the low-level code produced when a compiler translates high-level code.

12. All Java programming statements must end with a _____.

    a. period

    b. comma

    c. closing parenthesis

    d. semicolon

    Answer: d

    Feedback:

    All Java programming statements must end with a semicolon.

13. Arguments to methods always appear within _____.

    a. parentheses

    b. double quotation marks

    c. single quotation marks

    d. curly braces

    Answer: a

    Feedback:

    Arguments to methods always appear within parentheses.

14. In a Java program, you must use _____ to separate classes, objects, and methods.

    a. commas

    b. semicolons

    c. dots

    d. forward slashes

    Answer: c

    Feedback:

    In a Java program, you must use dots to separate classes, objects, and methods.

15. All Java applications must have a method named _____.

    a. `method()`

    b. `main()`

    c. `java()`

    d. `Hello()`

    Answer: b

    Feedback:

    All Java applications must have a method named `main()`. Although an application could have a method named `method()`, `java()`, or `Hello()`, they are not required method names. `Hello()` also is not a conventional name because methods usually start with a lowercase letter.

16. Nonexecuting program statements that provide documentation are called _____.

    a. classes

    b. notes

    c. comments

    d. commands

    Answer: c

    Feedback:

    Nonexecuting program statements that provide documentation are called comments.

17. Java supports three types of comments: _____, _____, and Javadoc.

    a. line, block

    b. string, literal

    c. constant, variable

    d. single, multiple

    Answer: a

    Feedback:

    Java supports line, block, and Javadoc comments.

18. Which of the following is not necessary to do before you can run a Java program?

    a. coding

    b. compiling

    c. saving

    d. debugging

    Answer: d

    Feedback:

    Although you should debug a program, it can run with bugs—it simply will produce incorrect results. You must code, compile, and save a program before you can run it.

19. The command to execute a compiled Java application is _____.

    a. `run`

    b. `execute`

    c. `javac`

    d. `java`

    Answer: d

    Feedback:

    The command to execute a compiled Java application is `java`. The command to compile a program is `javac`.

20. You save text files containing Java source code using the file extension _____.

    a. *.java*

    b. *.class*

    c. *.txt*

    d. *.src*

    Answer: a

    Feedback:

    You save text files containing Java source code using the file extension .java.